Transparent Base Rates Approach the State of the Art in Conflict Forecasting

Evidence from a fully auditable forecasting system, five preregistered covariate failures, and equal-weight forecast pooling

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Abstract

Machine-learning systems now define the state of the art in armed-conflict forecasting, but their opacity makes them difficult to audit, and a recurring finding—that naive baselines are nearly as accurate—has never been given a systematic, adversarially honest treatment by a system *designed* to be transparent. We present TOCSIN, a conflict-forecasting system whose every probability is a reference-class base rate: an empirically shrunk historical frequency, conditioned on recency structure, auditable down to the individual class counts. Evaluated walk-forward on the same target, vantages, and information sets as the Uppsala VIEWS system (7,680 country-months across five historical vantages), TOCSIN attains a Brier score of 0.0461 against VIEWS's 0.0412—within 12% of the state of the art, while beating it on 70% of individual country-months—and naive last-year persistence (0.0433) falls between them. We then subject the system to a preregistered tune/validate protocol and test five families of theoretically motivated covariates: regime type, forecast-horizon decay, youth bulges, militarized-dispute and alliance history, and ethnic exclusion (alone and interacted). All five fail out-of-sample, despite large descriptive associations (youth bulges quadruple onset rates; militarized-dispute history marks a 30-fold onset class among long-quiet country pairs). We identify three mechanisms—absorption by recency conditioning, class fragmentation under shrinkage, and, at rare-event base rates, a formal *metric blindness* of the Brier score, for which we give an exact decomposition showing that even perfect conditioning cannot move aggregate Brier by more than 0.05% at pair-grain base rates. Finally, zero-parameter equal-weight pooling of VIEWS with persistence outperforms every individual model (0.0399, −3.3% versus VIEWS), while adding TOCSIN to that pool *worsens* it: the transparent system wins months but duplicates the pool's recency information rather than diversifying it. The results sharpen a deflationary view of conflict predictability—the practical ceiling on the standard occurrence target sits close to what recency alone delivers, and is most cheaply approached not by richer models but by pooling diverse ones.

Keywords: conflict forecasting; base rates; reference-class forecasting; Brier score; forecast combination; preregistration; UCDP; VIEWS

*Independent researcher. Email: `kara@soulstone.org`. Code (MIT) and the merged dataset (CC BY 4.0) are public at `https://github.com/KaraZajac/TOCSIN`; a continuously updating deployment, including all diagnostics reported here, is at `https://tocsin.karazajac.io`. A single command regenerates every number in this paper from primary sources. Archived at DOI 10.5281/zenodo.21634193.

1 Introduction

Armed-conflict research has taken a decisive predictive turn. Where Ward et al. (2010) diagnosed a field optimizing statistical significance at the expense of predictive power, there now exist standing forecast systems—most prominently the Uppsala Violence Early-Warning System (VIEWS; Hegre et al., 2019)—that publish monthly machine-learning forecasts of state-based violence for every country, and organized competitions that score them (Hegre et al., 2022; Vesco et al., 2022). The predictive turn, however, has produced a persistent embarrassment alongside its successes: simple baselines—last year's violence, the country's historical rate—come uncomfortably close to the most sophisticated ensembles (Cederman and Weidmann, 2017; Chadefaux, 2017). This gap between methodological ambition and marginal accuracy gain is usually reported in passing, as a caveat. It has rarely been made the *object* of study.

This paper does so, from an unusual direction. Rather than build a richer model and report its increment, we build the most disciplined possible version of the *simple* alternative—a pure reference-class base-rate engine in the tradition of clinical-versus-statistical prediction (Meehl, 1954; Dawes, 1979) and outside-view forecasting (Kahneman and Lovallo, 1993)—and ask three questions with it:

1. **How close does transparent get?** When every probability is a shrunk historical frequency whose class definition, member counts, and shrinkage weights are inspectable, how much accuracy is sacrificed relative to the covariate-rich state of the art, scored on the state of the art's own target?

2. **What do covariates actually add, out-of-sample?** The conflict literature proposes many structural correlates of onset—regime type (Goldstone et al., 2010), development (Fearon and Laitin, 2003; Collier and Hoeffler, 2004), youth bulges (Urdal, 2006), ethnic exclusion (Cederman et al., 2010), and dyadic dispute history (Maoz et al., 2019). We test five covariate families under a preregistered tune/validate protocol designed to make post-hoc rationalization impossible, and report all results, including—especially— the failures.

3. **Where do the remaining gains live?** If single-model improvements are exhausted, the forecast-combination literature (Bates and Granger, 1969; Clemen, 1989; Timmermann, 2006) predicts that *pooling* diverse forecasters should beat the best of them. We test zero-parameter equal-weight pools against every individual system, and use the pool's composition to diagnose what information each system actually contributes.

The answers, in brief: transparent gets within 12% of the state of the art (and naive persistence within 5%); all five covariate families fail out-of-sample despite large descriptive associations, for reasons we can mechanically identify; and an equal-weight pool of the ML system with naive persistence beats everything, while our own system—despite winning 70% of individual months—*subtracts* from that pool because its information is already inside it. Together these results locate the practical ceiling of the standard conflict-occurrence target close to what recency structure alone delivers, and suggest the field's cheapest remaining accuracy lies in combination and in target design rather than in additional structure.

A secondary contribution is infrastructural. The entire system—a merged, crosswalked, CC BY-licensed dataset spanning nine sources; the engine; the evaluation harness; a self-resolving forecast journal; and a public, continuously updating deployment—is open and reproducible. The preregistration machinery (§6.1) is, to our knowledge, the first covariate-adoption protocol for a conflict-forecasting system that mechanically enforces a single consultation of held-out data and the retirement of spent validation eras.

2 Related work

Conflict forecasting systems. The Political Instability Task Force established that parsimonious models with regime-type interactions could anticipate instability onsets years ahead (Goldstone et al., 2010). VIEWS (Hegre et al., 2019, 2022) is the current reference system: a monthly-updated ensemble over UCDP data producing, among other targets, the probability of $\geq$25 state-based battle deaths per country-month—the target we adopt for head-to-head comparison. Blair and Sambanis (2020) argue theory-structured models travel better out-of-sample; Muchlinski et al. (2016) document large in-sample gains from random forests that Colaresi and Mahmood (2017) show require careful out-of-sample discipline to interpret.

The baseline embarrassment. Cederman and Weidmann (2017) caution that conflict's rare-event structure and the dominance of history-dependence put low ceilings on achievable gains; Chadefaux (2017) surveys the same pattern. Our contribution is to *instrument* this claim: we reproduce it independently (persistence within 5% of VIEWS on 7,680 scored country-months), quantify the covariate increment under preregistration (≈ 0), and give an exact scoring-rule decomposition for why large class-rate signals can be invisible to aggregate Brier at rare-event base rates.

Statistical versus clinical prediction and the outside view. The engine is a deliberate descendant of the actuarial tradition— Meehl (1954), Dawes (1979), and the reference-class forecasting of Kahneman and Lovallo (1993). Its wager is that in a domain governed by sticky, recurrent processes, a disciplined outside view captures most extractable signal. The five covariate failures in §6 are, in effect, a measured defense of that wager.

Forecast combination. That combinations of diverse forecasts outperform their best member is among the most replicated findings in the forecasting literature, from Bates and Granger (1969) through Clemen (1989) and the M-competitions (Makridakis and Hibon, 2000). Equal weights are notoriously hard to beat (the "forecast combination puzzle"; Timmermann, 2006). Section 7 imports this machinery into the conflict domain with a zero-parameter design that forecloses fitting-based objections, and adds a compositional diagnostic: which member's *removal* improves the pool.

Scoring rules. We score with the Brier score (Brier, 1950), a strictly proper rule (Gneiting and Raftery, 2007), and report skill relative to climatology. Section 6.3 formalizes a limitation of unweighted proper scores under extreme class imbalance that connects to the literature on rare-event verification and weighted scoring (Ferro and Stephenson, 2011): propriety guarantees honest elicitation, not sensitivity to refinements of small-mass classes.

3 Data

All inputs are public. The system merges nine sources onto a common spine—Gleditsch–Ward (G-W) country code × year (Gleditsch and Ward, 1999)—with year-aware crosswalks and a validation gate; the merged product is redistributed under CC BY 4.0 as `tocsin-dataset-26.1` with a full codebook.

- **UCDP/PRIO** (release 26.1): the Armed Conflict Dataset and its dyadic version (Gleditsch et al., 2002; Davies et al., 2026), the Georeferenced Event Dataset (Sundberg and Melander, 2013) with monthly *candidate* updates, non-state and one-sided tables, and the termination/episode coding of Kreutz (2010). Events with UCDP placeholder actors are excluded from dyad grain; candidate files are deduplicated by event id with latest-version precedence.
- **Gleditsch–Ward** system membership and minimum distances (Gleditsch and Ward, 2001), defining state-system exposure and the $\leq$400 km proximity relation.
- **PRIO Battle Deaths v3.1** (Lacina and Gleditsch, 2005), extending state-based death series to 1946 for descriptive use.
- **Powell–Thyne coups** (Powell and Thyne, 2011): 1950–present attempts and outcomes.
- **V-Dem Regimes of the World** (Lührmann et al., 2018), via Our World in Data.
- **OWID population** (UN WPP/HYDE composite), carried forward to the data edge.
- **World Bank WDI** covariates (income, inflation, age structure, urbanization, infant mortality; CC BY 4.0).
- **Ethnic Power Relations 2021** (Vogt et al., 2015): politically excluded population shares.
- **Correlates of War**: dyadic militarized interstate disputes 4.03 (Maoz et al., 2019; Palmer et al., 2022), National Material Capabilities v7 (Singer et al., 1972), and Formal Alliances v4.1 (Gibler, 2009), with a unit-tested COW$\to$G-W crosswalk (unified Germany 255$\to$260, Yemen 679$\to$678, Serbia via the year-aware 345$\to$340 alias, and the Pacific microstates remapped simultaneously).

Two engineered universes matter below. The **country universe** counts main-system G-W members (12,703 country-years, 1946–2025). The **pair universe** operationalizes politically relevant pairs as proximity ($\leq$400 km) $\cup$ same-region $\cup$ P5-membership: 243,609 undirected pair-years, within which 98.2% of all UCDP state-based dyadic activity occurs—constructed *ex ante*, never by conditioning on observed conflict.

4 The engine

4.1 Estimand and reference classes

For a unit u (country, dyad, or pair), year t, and measure m (deaths above a threshold; UCDP activity; episode termination; coup attempt), the engine estimates $\Pr(m$ holds for u in $t)$ as a shrunk historical frequency over a reference class defined entirely by *recency structure* observable before t:

- **Active units** are banded by episode age (consecutive hit-years: 1, 2–3, 4–9, 10+) $\times$ last-year intensity (below/above the UCDP war line of 1,000 deaths; for coups, failed/successful).

Table 1: Walk-forward performance across the seven evaluation suites.

suite	n	base rate	Brier	skill
country, sb deaths $\geq$25	5,547	.167	.0411	+70.6%
country, all-type deaths $\geq$100	5,547	.160	.0366	+72.8%
country, UCDP activity	11,274	.170	.0495	+65.0%
dyad, UCDP activity	24,876	.115	.0405	+60.6%
dyad, episode termination	2,797	.234	.1668	+7.1%
pair, onset/activity	234,136	.0006	.0005	+19.6%
country, coup attempt	6,145	.019	.0197	+17.6%

- **Non-active units** are banded by time since last hit (*recent* 2–3, *dormant* 4–10, *cold* 11+/never), with a spatial-contagion flag (+nbr) when any $\leq$400 km neighbor was in state-based conflict the prior year.
- At month grain, active units additionally carry a **tempo band** (trailing hit-months 1–3, 4–8, 9–12), and windows that are not calendar-aligned are priced on a rolling monthly substrate whose bucket construction provably reduces to the annual engine's for January-aligned windows.

4.2 Estimation

Within bucket b at level $\ell \in \{\text{self}, \text{region}, \text{global}\}$, with k hits in n class-years, the class rate is the Jeffreys-smoothed frequency $\hat{p} = (k + \frac{1}{2})/(n+1)$. Levels are combined by empirical-Bayes shrinkage (Efron and Morris, 1975): the self-level posterior is

$$\tilde{p}_{\text{self}} \;=\; \frac{k_{\text{self}} + M\,\hat{p}_{\text{parent}}}{n_{\text{self}} + M}, \tag{1}$$

with the prior strength M moment-matched from the dispersion of class rates across units and clamped to $[5, 1000]$; regions falling below 30 class-years fall back to the global class. The headline probability is the deepest level with adequate support, and every output ships the full ladder—counts, rates, M, and posterior at each level—plus caveat notes (left-censoring at the substrate edge, window approximations, provisional-data flags).

4.3 Walk-forward evaluation

All evaluation is walk-forward: scoring year t uses classes built from data strictly before t (five-year burn-in), so every reported score is out-of-sample by construction. A class-signature memoization makes the full seven-suite backtest ($\approx$290,000 unit-year records) run in seconds, which is what renders the preregistration protocol of §6.1 practical. Table 1 reports current walk-forward performance (skill $= 1 - \text{Brier}/\text{Brier}_{\text{climatology}}$).

Two design decisions were themselves settled empirically and are kept as negative results: horizon-aware class decay (pricing a forecast g years past the data edge by class rates offset g) measured *worse* (+.003 Brier; class-level decay pools exiting with persisting units), so one-step rates are applied frozen; and a promote-only nowcast lets provisional current-year data upgrade but never downgrade a bucket.

Table 2: The arena: Brier scores on Pr(≥25 sb deaths per country-month), five vantages, horizons 1–12. Pools are equal-weight linear opinion pools (zero fitted parameters).

model	Brier	skill	h1–3	h4–6	h7–12
pool: VIEWS + persistence	**.03987**	+56.8%	.0340	.0376	.0439
pool: all three	.04087	+55.7%	.0354	.0382	.0450
VIEWS	.04123	+55.3%	.0337	.0396	.0458
persistence	.04326	+53.1%	.0389	.0401	.0471
TOCSIN	.04610	+50.0%	.0413	.0425	.0503
climatology	.09219	0	.0877	.0929	.0940

5 The arena: scoring against the state of the art

5.1 Design

We score all systems on VIEWS's native target—Pr(≥25 state-based battle deaths in a country-month), UCDP best estimate—at each of five historical vantages (VIEWS `fatalities002` runs of April and October 2023, April and October 2024, April 2025), horizons 1–12 months, using only information available at each vantage. TOCSIN runs walk-forward-clamped (classes end at the vantage). The VIEWS probability is its published `main_dich`, whose semantics we verified empirically rather than assumed: across pre-2024 runs its mean (.138) matches the realized ≥25 frequency (.136), not the ≥1 frequency (.251). Baselines: **persistence** (country's trailing-12-month hit rate, add-one smoothed) and **climatology** (full-history monthly rate, Jeffreys-smoothed). 7,680 scored country-months; 7% of outcomes are provisional (candidate GED) at scoring time.

5.2 Results

Table 2 reports the standings. Three readings, before the pools (deferred to §7):

1. **The transparency tax is ∼12%.** A pure lookup table—auditable to the class counts—concedes .0049 Brier to a covariate-rich ML ensemble on the ensemble's home target. Half of an initially larger gap was closed by a single recency-structure refinement (the tempo band moved TOCSIN from .0611 to .0461), i.e., by *better history*, not covariates.

2. **The baseline embarrassment replicates.** Persistence sits within 5% of VIEWS. This is an independent reproduction, on a fixed public target with verified semantics, of the pattern Cederman and Weidmann (2017) warned about.

3. **Aggregate and disaggregate disagree.** TOCSIN produces the better forecast in 5,386 of 7,657 decided country-months (70%) yet loses on aggregate: it wins small on the quiet mass and pays large on transitions, the fingerprint of a system whose errors concentrate exactly where VIEWS's covariates earn their keep. The horizon decomposition agrees: the deficit is largest at h1–3 (.0413 vs .0337), where fresh signal matters most, and smallest at h7–12 (.0503 vs .0458).

We refrain from formal tests of Brier differences: the 7,680 records are strongly dependent (overlapping 12-month horizons from five vantages; within-country serial correlation), and a clustered Diebold–Mariano treatment on five effective vantages would be underpowered theater. The differences we interpret—pools versus singles, and the covariate nulls below—are either zero-parameter comparisons or preregistered decisions, not significance claims.

6 The covariate program: five preregistered failures

6.1 The tune/validate protocol

Walk-forward evaluation removes look-ahead but not *designer overfitting*: a researcher who iterates conditioning schemes against the full scored history is fitting to it. After one such episode—a V-Dem regime conditioning that looked plausible and measured worse on every country suite (e.g., sb≥25 .0411→.0413; all-type≥100 .0366→.0371)—we froze a protocol:

1. Split scored years at 2007: **tune** (≤2007), **validate** (>2007).
2. Candidate schemes are enumerated *a priori*; any thresholds are computed from tune-era data only.
3. Candidates compete on tune; the winner is compared to baseline on validate **exactly once**.
4. Adoption requires ≥1% relative Brier improvement on validate *and* a consistency guard on tune (a majority of cuts helping, for parametric families; the winner itself beating baseline, for heterogeneous sets).
5. Every consultation of validate is logged; when an era's validate set has been consulted for a family of hypotheses, it is **retired**— subsequent studies require genuinely fresh outcome years.

The 1% bar was set before any study ran, calibrated to the observed noise scale of the validate split (a scheme that lost on tune still swung +0.4% on validate; see below). The protocol's output is a verdict, not a p-value; its epistemic content is that validate data cannot be shopped.

6.2 Results

Study 1 — youth bulges (country grain). Descriptively, age structure is the strongest single onset covariate we measured: among 6,280 currently-quiet country-years with covariate coverage (base onset rate 3.3%), those with >39% of population under 14 onset at 6.5% versus 1.7% for older populations—a ∼4× lift, consistent with Urdal (2006). Under the protocol (candidate cuts at the 50th/60th/66th/75th tune-era percentiles): **zero of four cuts beat the baseline engine on tune** (best .05384 vs .0537); the least-bad cut carried a +0.4% relative edge to validate—below the bar, and with the tune guard failed. **Rejected.** A naive adoption threshold (ΔBrier $< -.0001$) would have adopted a covariate that could not beat its own tuning baseline; the guard exists precisely for this case.

Study 2 — dispute and alliance history (pair grain). Four candidates on the pair universe: any militarized-dispute history (ever-MID), MID within 25 years, fatal-MID history, and a defense-pact flag (COW alliances, last observed status carried forward). Descriptively the leading candidate is enormous: among pair-years entering the *cold* bucket, those with MID history onset at 0.194% (44/22,649) versus 0.006% (14/215,855) without—a **30**× class-rate ratio. Under the protocol: **zero of four candidates beat baseline on tune** (.000574 vs .000573); the tune-selected candidate (ever-MID) was 0.7% relatively *worse* on validate. **Rejected**—for a reason §6.3 shows is structural rather than empirical.

Study 3 — exclusion and the joint hypothesis (country grain; final consultation of the 2007 era). The one descriptively motivated interaction not yet tested: ethnic exclusion alone, youth∧exclusion (the compound cell onsets at 8.6% versus 2.9% for older/low-exclusion years), and youth∨exclusion—deliberately two-cell flags rather than fragmenting cross-products. **Zero of three beat baseline on tune**; the tune-selected candidate (exclusion alone) was 1.0% relatively worse on validate—the worst held-out reading of any study. **Rejected**, and the 2007 validation era was retired as preregistered (country validate consulted twice, pair once).

Together with the two pre-protocol ablations (regime conditioning; horizon decay), the scoreboard is **five families proposed, five rejected**, while the engine's own recency refinements (episode age, intensity, tempo, neighbor) were adopted on the same walk-forward evidence. The pattern is not that structural covariates are uninformative—descriptively they are strong—but that they are informative *about the same thing recency already measures.* A young, exclusionary state that has been at peace for thirty years is, empirically, at low risk; a middle-aged democracy one year out of civil war is not. Conflict history is not a proxy to be improved upon; it is the sufficient statistic the covariates approximate.

Two mechanisms recur. **Absorption**: conditional on bucket, the covariate's class-rate difference shrinks toward zero (youth's 4× marginal lift is mostly the young-countries-fight-recently composition effect). **Fragmentation under shrinkage**: splitting a bucket cuts class support; moment-matched shrinkage then pulls both children toward the parent, so even a real difference buys little resolution while adding estimator variance—the measured regime failure, and the reason the joint study used two-cell flags. The third mechanism deserves its own subsection.

6.3 Metric blindness at rare-event base rates

The pair-grain rejection is *not* evidence that dispute history is uninformative—the 30× class ratio is real and walk-forward. It is evidence about the scoring rule. Decompose the maximum possible Brier improvement from any refinement of a class with pooled rate $\bar{p}$ into subclasses c with masses w_c and true rates π_c (this is the resolution term of Murphy's decomposition; Murphy, 1973):

$$\Delta\text{Brier}_{\max} \;=\; \sum_{c} w_c\,(\pi_c - \bar{p})^2. \tag{2}$$

For the cold-pair class (238,504 pair-years entering *cold*): $w_1 = 0.095$, $\pi_1 = .00194$; $w_0 = 0.905$, $\pi_0 = .000065$; $\bar{p} = .000243$. Then $\Delta\text{Brier}_{\max} = 0.095 \cdot (.00170)^2 + 0.905 \cdot (.00018)^2 \approx 3.0 \times 10^{-7}$—**0.05% of the .000573 tune-era baseline**, twenty times below the preregistered 1% bar. At these base rates, *perfect* conditioning is invisible to aggregate Brier: the quiet mass is already scored nearly perfectly, and the flagged class is too small for its (large, real) improvement to register. The same arithmetic explains why the engine's 7× neighbor-contagion signal barely moves month-grain Brier, and why pair-grain skill (+19.6%) looks modest despite genuine discrimination.

The methodological point generalizes: **strictly proper scores guarantee honest elicitation, not sensitivity to policy-relevant refinements under extreme class imbalance.** A field that selects covariates by aggregate Brier on rare-event targets will systematically discard

Table 3: Equal-weight pools versus the best single model.

pool	Brier	vs best single
VIEWS + persistence	**.03987**	**−3.3%**
VIEWS + persistence + TOCSIN	.04087	−0.9%
VIEWS + TOCSIN	.04144	+0.5%
persistence + TOCSIN	.04329	+5.0%

its strongest onset signals. Candidate remedies—weighted proper scores (Ferro and Stephenson, 2011), the log score (which diverges precisely where Brier saturates), or class-conditional evaluation—must be *preregistered*, because switching metrics after observing results reintroduces the shopping the protocol exists to prevent. In our own system the verdict stands (the era is spent); the 30× signal ships as displayed context on pair forecasts, labeled as having failed the adoption bar. A log-loss bar for the pair grain is preregistered here for the next evaluation era (outcome years ≥2026).

7 Pooling: the best forecast is nobody's model

The head-to-head structure of §5—TOCSIN better on 70% of months, worse on aggregate—is the textbook precondition for combination gains: the systems' errors are differently shaped. We therefore score equal-weight linear opinion pools. Equal weights involve zero fitted parameters, so scoring them on the same vantages is measurement, not tuning; the adoption criterion (beat the best member) was fixed in advance, and all four candidate pools are reported (Table 3).

Two results. First, the field's cheapest available accuracy gain on this target is not a model at all: averaging the ML system with naive persistence beats everything, at every horizon band, for free. This is the Bates–Granger result landing in the conflict domain with unusual force, and it should discipline how single-model improvements are marketed: a new system that does not beat the *pool* has not advanced the frontier.

Second, the pool is a diagnostic instrument. Adding TOCSIN to VIEWS+persistence *worsens* it (.03987→.04087) despite TOCSIN's 70% month-wise win rate, and TOCSIN+persistence is barely better than persistence alone. The inference is sharp: at month grain, TOCSIN's forecast is (approximately) a recalibrated persistence—its information is already inside the pool, so it adds redundancy, not diversity. Pools pay for *orthogonal* error, and VIEWS's covariate machinery is the orthogonal ingredient. This reframes the transparency tax of §5: the transparent system's residual value on this target is not incremental accuracy (persistence supplies its signal more cheaply) but auditability—and its *distinct* value lies on targets no other system prices, to which we return below.

8 Discussion

A deflationary synthesis. Assembled, the results argue that the standard country-month occurrence target is close to its practical predictability ceiling: (i) a pure recency machine reaches within 12% of the state of the art; (ii) naive persistence reaches within 5%; (iii) five theoretically central covariate families add nothing out-of-sample once recency is conditioned on; (iv) the cheapest remaining gain is a zero-parameter average. None of this says conflict is unpredictable—skill versus climatology is +50–57% throughout—it says the *predictable part is mostly history*, and that the marginal covariate, however strong its bivariate association, is purchasing information the forecast already owns. This is the caution of Cederman and Weidmann (2017), converted from warning into measurement.

What transparency bought. The auditable design is not (on this target) an accuracy strategy; it is an epistemics strategy. Because every probability decomposes into named classes and counts, failures localize: we can state *why* youth fails (absorption), why regime fails (fragmentation), why dispute history fails (metric blindness)—mechanisms, not shrugs. The same architecture made the preregistration protocol cheap enough to actually follow, and the negative results reportable at publication grade. We suggest the field's forecasting systems adopt adoption-protocols of this shape: enumerated candidates, tune/validate separation, single consultation, retirement of spent eras, and published rejections.

Rare events need preregistered metrics. The decomposition of §6.3 implies that aggregate Brier is the wrong adoption criterion wherever base rates are far from the forecast probabilities' working range—dyadic onset, interstate war, mass-atrocity onset. Weighted or logarithmic criteria see what Brier cannot, but only preregistration keeps them honest.

Limitations. (1) The arena scores one target at one grain—VIEWS's home turf; TOCSIN's design center (annual horizons, arbitrary thresholds, termination, coups—the operational questions in its journal) has no external comparator yet, and our claims are correspondingly scoped. (2) Five vantages with overlapping horizons preclude meaningful significance testing on Brier differences; our decisive comparisons are zero-parameter or preregistered instead. (3) Seven percent of arena outcomes were provisional (candidate GED) at scoring time; UCDP revisions could shift third-decimal values. (4) VIEWS is represented by its published dichotomous output under empirically verified semantics; other VIEWS surfaces (expected fatalities) are not evaluated. (5) The covariate verdicts are specific to this engine's conditioning structure and adoption bar; they bound the *increment over recency*, not the covariates' scientific relevance. (6) The 2007-split era is spent; our own future covariate work must wait for fresh outcome years, and we commit here to the preregistered log-loss bar for the pair grain.

Future work. The deployed system accumulates the missing evidence automatically: a journal of operational forecasts resolves monthly against UCDP updates, building both an annual-grain arena on TOCSIN's own targets and the fresh validation era (outcomes $\geq$2026) that the next covariate—a continuous tempo input aimed at the measured h1–3 deficit—must clear. The combination result suggests a standing published pool alongside; the metric result suggests the community converge on preregistered rare-event scoring before, not after, the next generation of systems reports its gains.

9 Reproducibility

Everything is public. Code (MIT) and the merged dataset (CC BY 4.0, with per-source attribution chains and codebook) are at `https://github.com/KaraZajac/TOCSIN`; the live system, including the arena table, all walk-forward calibration diagnostics, and every forecast with its full reference-class ladder, is at `https://tocsin.karazajac.io`. A single command (`tocsin` `pull && tocsin build && tocsin backtest && tocsin benchmark`) regenerates every number in this paper from primary sources; the preregistration protocol is `tocsin protocol`. The five-vantage arena pins the exact VIEWS runs consumed. Raw ACLED data informs a display-only divergence indicator on the site and is neither redistributed nor used in any result reported here.

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